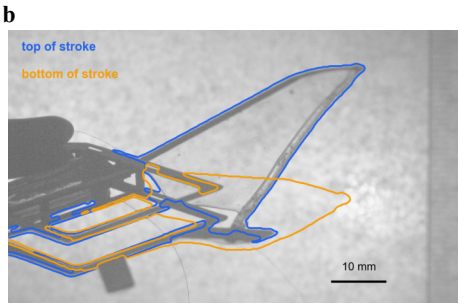
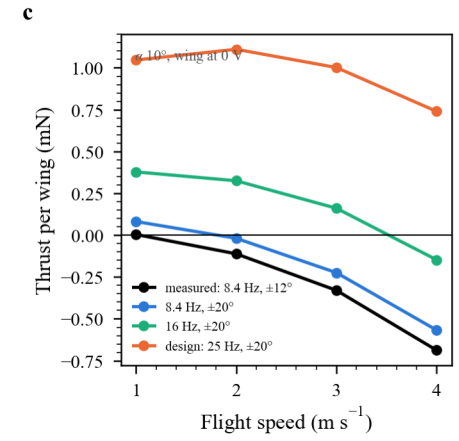
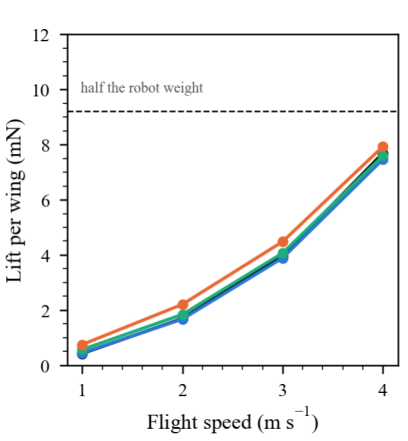
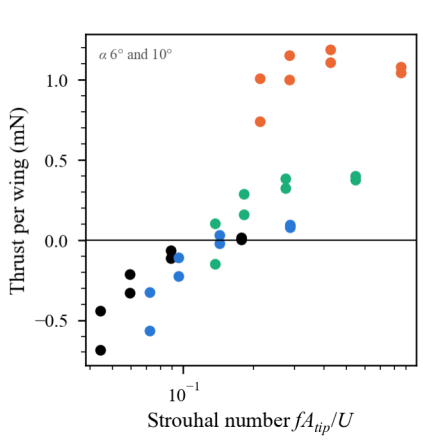

a
as built: 8.4 Hz, ±12°
net drag
16–25 Hz, ±20°
net thrust
St = f A / U
robot now, 0.05
thrust (animals cruise at 0.2–0.4)
0
0.1
0.2
0.3
0.4


b
top of stroke
bottom of stroke
10 mm


c
α 10°, wing at 0 V
Thrust per wing (mN)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
measured: 8.4 Hz, ±12°
8.4 Hz, ±20°
16 Hz, ±20°
design: 25 Hz, ±20°
1
2
3
4
Flight speed (m s^{-1})


d
half the robot weight
Lift per wing (mN)
12
10
8
6
4
2
0
1
2
3
4
Flight speed (m s^{-1})


e
α 6° and 10°
Thrust per wing (mN)
1.0
0.5
0.0
−0.5
10^{-1}
Strouhal number fA_{tip}/U